

How Much Can Recursive AI Self-Improvement Raise Economic Growth?

Oliver Pardo

September 2026

Abstract

I study how recursive AI self-improvement affects long-run economic growth in a decentralized economy with autonomous AI research. A monopolistic developer supplies AI services and invests in AI efficiency without a finite efficiency frontier. With unit elasticity of substitution between AI services and labor, I derive a balanced-growth equilibrium in which research spending, saving, and prices are endogenous. The increase in long-run growth depends on research returns, the production weight of AI, and exogenous effective-labor growth. Output per person and real wages grow at the same rate, while labor retains a positive income share. I use this analytical benchmark to organize calibration and parameter sensitivity. The inherited illustrative calibration implies a growth increase of about 0.087 percentage points per year; this is a conditional model calculation, not an empirical estimate or a forecast.

1 Introduction

How much can economic growth increase if AI systems help develop better AI? Answering this question requires connecting the productivity of AI research to the allocation of resources in the rest of the economy. Research competes with current consumption, physical investment, and the compute used to supply AI services. Its scale depends on the profits that improvements can earn and on the interest rate at which those profits are discounted.

I study these decisions in an endogenous-growth model with a representative household, competitive final-good firms, and a monopolistic AI developer. Final production combines capital, labor, and AI services. The developer uses AI research services to raise the efficiency with which compute produces both production and research services. Research is autonomous: the model describes an economy in which human researchers are no longer a necessary current input into AI development. It does not describe how that transition to autonomous research occurs.

This paper develops the uncapped unit-elastic benchmark in Appendix C of [Pardo \(2026\)](#). That paper compares substitution regimes under a finite AI-efficiency frontier. Here I fix the elasticity of substitution at one and remove that frontier. I derive an equilibrium with constant growth rates and endogenous allocation shares, rather than assume fixed saving and research expenditure. This restriction isolates a tractable quantitative question; it is not a claim that aggregate AI–labor substitution has been estimated to equal one.

I distinguish inherited macroeconomic reference values from uncertain AI parameters. The analytical results identify which parameters affect permanent growth and which affect reference levels.

This first draft establishes that benchmark and the calibration framework. Empirically disciplined ranges for the AI parameters and quantitative sensitivity exercises remain to be developed. The calculations do not assume that the present world economy lies on the model's balanced-growth path.

2 Model

Time is continuous and measured in years. Population N and labor-augmenting technology A satisfy

$$\dot{N} = nN, \quad \dot{A} = \gamma A, \quad n \geq 0, \quad \gamma > 0. \quad (1)$$

Each person supplies one unit of labor inelastically. Production labor is denoted by L ; labor-market clearing will impose $L = N$. For any positive variable V , write $g_V = \dot{V}/V$. The final good is the numeraire.

2.1 Household

The household owns physical capital K and the AI developer. Let C denote aggregate consumption, w the real wage, r the net real interest rate, and Π the developer's net distribution to its owner. Given their paths and $K_0 > 0$, the household chooses consumption to solve

$$\max \int_0^\infty e^{-\rho t} N_t \log(C_t/N_t) dt, \quad \rho > n, \quad (2)$$

subject to

$$\dot{K} = rK + wN + \Pi - C, \quad C > 0, \quad K \geq 0. \quad (3)$$

The parameter ρ is the discount rate. For an interior optimum, the Euler and transversality conditions are

$$g_C = n + r - \rho, \quad (4)$$

$$\lim_{t \rightarrow \infty} e^{-\rho t} \frac{N_t K_t}{C_t} = 0. \quad (5)$$

Depreciation is already deducted from r .

2.2 Final-good firms

Let X denote AI production services. I set the elasticity of substitution between effective labor AL and X to $\sigma = 1$, retaining the production weights $\omega_X \in (0, 1)$ and $\omega_L = 1 - \omega_X$. With capital elasticity $\alpha \in (0, 1)$, production is

$$Y = K^\alpha [(AL)^{\omega_L} X^{\omega_X}]^{1-\alpha}. \quad (6)$$

Define the output elasticities

$$\beta = (1 - \alpha)\omega_X, \quad \lambda = (1 - \alpha)\omega_L, \quad \alpha + \beta + \lambda = 1. \quad (7)$$

Thus the same production function is $Y = K^\alpha(AL)^\lambda X^\beta$. Let p_X denote the AI-service price and $\delta \geq 0$ capital depreciation. A competitive final-good firm solves

$$\max_{K,L,X \geq 0} \{K^\alpha(AL)^\lambda X^\beta - (r + \delta)K - wL - p_X X\}. \quad (8)$$

Its interior first-order conditions are

$$r = \alpha Y/K - \delta, \quad (9)$$

$$w = \lambda Y/L, \quad (10)$$

$$p_X = \beta Y/X. \quad (11)$$

Constant returns imply zero final-good profit. At given K, A, L , equation (11) defines inverse demand $p_X(X)$ for the developer.

2.3 The AI developer

Inference compute U and research compute M are flows of computational resources, each costing one unit of the final good per unit of compute. They are rival inputs, not stocks of processors. AI efficiency B converts them into the flows BU of AI production services and BM of AI research services. The same nonrival efficiency index augments both uses of compute. The service technology and research law are

$$X = BU, \quad (12)$$

$$\dot{B} = \chi(BM)^\eta, \quad \chi > 0, \quad 0 < \eta < 1. \quad (13)$$

The parameter χ shifts research productivity; η is the elasticity of the efficiency increment $\dot{B}$ with respect to research services BM . Efficiency has no finite upper bound. Higher B raises research productivity at a given M , creating recursive self-improvement. There is no human research input in this benchmark.

The developer exclusively owns B , sells X , and purchases $U = X/B$ and M . Ownership makes some benefits of research appropriable through monopoly rents. Taking K, A, L, r as given and starting from $B_0 > 0$, it maximizes

$$\int_0^\infty D_t [p_X(X_t)X_t - X_t/B_t - M_t] dt, \quad D_t = \exp\left(-\int_0^t r_s ds\right), \quad (14)$$

subject to (13), $X, M \geq 0$, and finite feasible allocations at every finite date. Since X does not

enter the research law directly, define optimized operating profit, before research expenditure, by

$$\pi(B) = \max_{X \geq 0} \{p_X(X)X - X/B\}. \quad (15)$$

Dependence on current K, A, L is implicit. Revenue is proportional to X^β , so the objective is strictly concave, its derivative is positive near zero, and linear cost dominates revenue at infinity. The unique positive optimum therefore satisfies

$$\beta p_X = 1/B. \quad (16)$$

Combining this condition with (11) and (12), and applying the envelope theorem, gives

$$p_X X = \beta Y, \quad U = \beta^2 Y, \quad \pi_B = U/B. \quad (17)$$

Let q be the current-value shadow price of AI efficiency. The Hamiltonian is $\pi(B) - M + q\chi(BM)^\eta$. Interior research and the costate condition give

$$1 = q\chi\eta B^\eta M^{\eta-1}, \quad (18)$$

$$\dot{q} = rq - U/B - \eta q \dot{B}/B. \quad (19)$$

Research equates its unit resource cost to the value of the efficiency it creates. Holding efficiency also yields compute-cost savings U/B and raises the productivity of subsequent research. The required return rq equals those two benefits plus the capital gain $\dot{q}$. The developer's transversality condition is

$$\lim_{t \rightarrow \infty} D_t q_t B_t = 0. \quad (20)$$

Net distributions are $\Pi = \pi(B) - M = p_X X - U - M$ and can be negative.

2.4 Equilibrium

Definition 1. *Given parameters and initial conditions $A_0, N_0, K_0, B_0 > 0$, an interior equilibrium is a trajectory for $(K, B, C, q, Y, L, X, U, M, r, w, p_X, \Pi)$, finite at each finite date, with $K, B, C, q, Y, L, X, U, M > 0$, such that the household and developer attain their global optima, final-good firms maximize profit, and markets clear. Its equations and boundary conditions are*

$$A_t = A_0 e^{\gamma t}, \quad N_t = N_0 e^{nt}, \quad L_t = N_t, \quad (21)$$

$$Y = K^\alpha (AL)^\lambda X^\beta, \quad (22)$$

$$X = BU, \quad (23)$$

$$p_X = \beta Y/X, \quad \beta p_X = 1/B, \quad (24)$$

$$r = \alpha Y/K - \delta, \quad w = \lambda Y/L, \quad (25)$$

$$\Pi = p_X X - U - M, \quad (26)$$

$$\dot{K} = Y - C - U - M - \delta K, \quad (27)$$

$$\dot{B} = \chi(BM)^\eta, \quad (28)$$

$$\dot{C} = (n + r - \rho)C, \quad (29)$$

$$1 = q\chi\eta B^\eta M^{\eta-1}, \quad (30)$$

$$\dot{q} = rq - U/B - \eta q \dot{B}/B, \quad (31)$$

$$K(0) = K_0, \quad B(0) = B_0, \quad (32)$$

$$\lim_{t \rightarrow \infty} e^{-\rho t} N_t K_t / C_t = 0, \quad (33)$$

$$\lim_{t \rightarrow \infty} D_t q_t B_t = 0. \quad (34)$$

Capital accumulation is the aggregate resource constraint; the household budget follows from it and the factor-payment and distribution equations. The initial jump variables C_0, q_0 are endogenous. This definition imposes no balanced-growth restrictions. The first-order conditions and TVCs alone are not asserted to be sufficient for equilibrium: global optimality is verified under explicit conditions below.

3 Balanced growth and equilibrium existence

Substituting $X = B\beta^2 Y$ from (17) into (6) yields

$$Y^{1-\beta} = K^\alpha (AL)^\lambda (\beta^2 B)^\beta. \quad (35)$$

Consider a candidate balanced-growth path (BGP) with positive constant $K/Y, C/Y, M/Y$. Differentiating (35) and the research law (13) gives, respectively,

$$g_Y = n + \gamma + (\beta/\lambda)g_B, \quad (36)$$

$$(1 - \eta)g_B = \eta g_Y. \quad (37)$$

The first equation relates AI improvement to aggregate production. The second requires research compute to grow with output when its expenditure share is constant. Define

$$\Delta = \lambda(1 - \eta) - \beta\eta = (1 - \alpha)(1 - \eta - \omega_X). \quad (38)$$

If $\Delta > 0$, the two growth equations have the solution

$$g_Y^* = \frac{(1 - \eta)(1 - \omega_X)}{1 - \eta - \omega_X} (n + \gamma), \quad (39)$$

$$g_B^* = \frac{\eta(1 - \omega_X)}{1 - \eta - \omega_X} (n + \gamma). \quad (40)$$

The superscript $*$ identifies this BGP. The wage equation and household Euler condition then imply

$$g_{Y/N}^* = g_w^* = \gamma + \frac{\eta\omega_X}{1 - \eta - \omega_X}(n + \gamma), \quad (41)$$

$$r^* = \rho + g_{Y/N}^*. \quad (42)$$

To check that these rates belong to an equilibrium, define the candidate ratios $k^* = K/Y$, $u^* = U/Y$, $i^* = (\dot{K} + \delta K)/Y$, $m^* = M/Y$, and $c^* = C/Y$. These are capital–output and expenditure ratios, not quantities per unit of effective labor. Their values are

$$k^* = \frac{\alpha}{r^* + \delta}, \quad u^* = \beta^2, \quad (43)$$

$$i^* = (g_Y^* + \delta)k^*, \quad (44)$$

$$m^* = \frac{\beta^2 \eta g_B^*}{\rho - n + \eta g_Y^*}, \quad (45)$$

$$c^* = 1 - u^* - i^* - m^*. \quad (46)$$

The research ratio follows from the developer’s conditions, not a fixed research rule. Indeed, (18) implies $M = \eta q g_B B$. Constant M/Y then gives $g_q = g_Y - g_B$. Dividing (19) by q and substituting these expressions yields (45). The remaining ratios follow from the capital return, the static monopoly solution, and resource clearing.

Proposition 1 (A balanced-growth equilibrium). *Maintain the primitive restrictions in Section 2. Suppose, in addition, that*

$$0 < \eta < \alpha, \quad \eta \leq \frac{1}{2}, \quad \beta \leq \frac{1}{2}, \quad \eta + \omega_X < 1. \quad (47)$$

For every $A_0, N_0 > 0$, there exist $K_0^, B_0^* > 0$ and an equilibrium with the rates and ratios in (39)–(46). Consumption is positive, both agents attain finite-valued global optima, and both transversality conditions hold. AI efficiency grows without bound as time tends to infinity but remains finite at every finite date.*

Proof. See Appendix A, Proof of Proposition 1. □

The restrictions serve different purposes. The condition $\eta + \omega_X < 1$ is necessary for this positive finite-rate BGP when $n + \gamma > 0$. If it fails, the two growth restrictions either contradict one another or require declining AI efficiency. This excludes the specified BGP, not every possible equilibrium trajectory. The bounds $\beta \leq 1/2$ and $\eta \leq 1/2$ are sufficient concavity conditions for global developer optimality. They are not claimed to be necessary. I retain $\eta < \alpha$ for comparability with the companion framework in Pardo (2026); the proof below does not separately require that comparison.

Proposition 1 constructs the initial stocks together with the BGP. It does not prove equilibrium existence for arbitrary fixed K_0, B_0 . Appendix C of Pardo (2026) also establishes local equilibrium

trajectories from nearby initial stocks under explicit stability and state-projection conditions. No off-BGP simulation is presented here.

The BGP income shares are

$$\frac{wL}{Y} = \lambda, \quad \frac{(r + \delta)K}{Y} = \alpha, \quad \frac{p_X X}{Y} = \beta, \quad \frac{\Pi}{Y} = \beta - \beta^2 - m^*. \quad (48)$$

Real wages and output per person grow together faster than γ while labor retains a positive income share. This differs from the characterized unit-elastic finite-frontier equilibria, where both growth rates tend to γ . Removing the frontier changes the long-run result; the uncapped BGP is not obtained by interchanging the long-run and infinite-frontier limits.

4 Calibration and measurement

Table 1 imports the reference values documented in the quantitative section of [Pardo \(2026\)](#). It separates macroeconomic benchmarks from illustrative AI parameters. Reusing these values preserves comparability; it does not make them empirical estimates of this autonomous-research economy.

Table 1: Inherited parameters and their evidential status

Parameter	Reference	Interpretation	Source or status
α	0.33	Capital elasticity	Inherited one-third macroeconomic benchmark.
δ	0.05	Annual depreciation	Inherited benchmark; PWT is the comparison source, not a direct estimate of this value (Feenstra et al., 2025).
ρ	0.04	Annual discount rate	Preference calibration, not a measured market interest rate.
n	0.003	Annual population growth	Inherited constant-rate approximation to the UN 2024–2100 world projection (United Nations, 2024).
γ	0.01	Annual labor-augmenting progress	Illustrative trend, with the OECD long-run scenarios as a scale comparison (OECD, 2025).
ω_X	0.20	AI production weight	Illustrative, not estimated. Not the AI revenue share itself.
ω_L	0.80	Effective-labor weight	Defined by $1 - \omega_X$.
η	0.20	Research-services elasticity	Illustrative, not estimated.
σ	1	AI–labor substitution	Maintained restriction of this paper.
χ	Free, > 0	Research productivity	Does not affect the BGP rates or expenditure ratios. Needs units and level targets for a quantitative level comparison.

The population parameter extrapolates a finite-period approximation, not a UN prediction of positive population growth forever. Likewise, measured labor-productivity growth is not the same object as the growth of A in this model. The OECD comparison does not directly identify γ . Both

n and γ therefore belong in long-run sensitivity analysis.

The finite-frontier paper uses $\chi = 1.4378$ to target an illustrative transition duration. I do not transfer that timing target to the uncapped economy. Equations (39)–(46) are independent of χ , whereas the initial BGP levels constructed in Appendix A depend on it. Calibrating χ cannot select an arbitrary permanent growth rate.

At the inherited reference values, the analytical calculations give

$$g_{Y/N}^* = g_w^* \simeq 1.0867\% \text{ per year}, \quad r^* \simeq 5.0867\% \text{ per year}. \quad (49)$$

The growth increase relative to $\gamma = 1\%$ is about 0.0867 percentage points per year. This comparison is between long-run model benchmarks, not a transition from today's economy and not an estimate of AI's observed impact.

Empirical discipline requires a mapping from observables to model objects. At unit elasticity, (48) implies that a measured AI revenue share would map to ω_X only after dividing by $1 - \alpha$, provided the measurement matches the model's industry boundary and pricing assumptions. AI investment, compute expenditure, and AI revenue are different objects. Measurement difficulties discussed by Fonteneau et al. (2025) motivate caution; they do not select $\omega_X = 0.20$.

For research, taking logs of (13) gives

$$\log \dot{B} = \log \chi + \eta \log B + \eta \log M. \quad (50)$$

The left side is the log of an efficiency increment, not the growth rate g_B . Moreover, research compute is an endogenous choice. Historical algorithmic improvements cannot be read directly as an estimate of η : they may also reflect human research, data, and complementary inputs omitted from the autonomous benchmark. Identifying defensible ranges for η and ω_X remains a separate empirical task.

5 Sensitivity and the quantitative question

The analytical sensitivity follows directly from (41). Within the maintained existence region,

$$\frac{\partial g_{Y/N}^*}{\partial \eta} = \frac{\omega_X(1 - \omega_X)}{(1 - \eta - \omega_X)^2}(n + \gamma) > 0, \quad (51)$$

$$\frac{\partial g_{Y/N}^*}{\partial \omega_X} = \frac{\eta(1 - \eta)}{(1 - \eta - \omega_X)^2}(n + \gamma) > 0. \quad (52)$$

The same derivatives apply to real-wage growth and to the interest rate when ρ is fixed. These are comparisons of equilibria, not partial adjustments holding research expenditure fixed.

The growth increase is amplified as $\eta + \omega_X$ approaches one from below. This makes the empirical plausibility of the parameter combinations particularly important. Such sensitivity is not itself evidence that large growth increases are likely. Parameter values outside the sufficient conditions of

Proposition 1 must not be presented as equilibria merely because the closed-form growth expression can be evaluated.

The next quantitative exercise will vary the two AI parameters jointly and report output-per-person growth, wage growth, the interest rate, and the research expenditure share. Every configuration must satisfy the stated existence restrictions. Sensitivity to n and γ will distinguish assumptions about AI from assumptions about exogenous effective-labor growth. Changes in α, δ, ρ also matter for allocations: their absence from the growth premium does not make them irrelevant to the equilibrium.

The no-AI growth benchmark is γ . Comparing the positive-AI BGP with that benchmark combines the production and recursive-research channels. It does not isolate the causal contribution of recursion. A counterfactual that retains AI services in production while removing their contribution to research productivity requires a separately specified and solved research technology; it is not implemented in this draft.

The current contribution is an analytical, globally verified BGP benchmark and a transparent starting calibration. The magnitude implied by the reference values is not a forecast. The next step is to discipline the uncertain AI parameters with evidence and then report conditional ranges, without treating sensitivity scenarios as estimated probabilities.

A Proofs

Proof of Proposition 1. First, all candidate ratios are positive. The denominator in (45) is positive, and

$$i^* = \frac{\alpha(g_Y^* + \delta)}{\rho - n + g_Y^* + \delta} < \alpha, \quad m^* < \frac{\beta^2 \eta}{1 - \eta}.$$

Since $\Delta = (1 - \alpha)(1 - \eta) - \beta > 0$, both $\beta < 1 - \eta$ and $\beta < 1 - \alpha$. Hence

$$u^* + m^* < \frac{\beta^2}{1 - \eta} < \beta < 1 - \alpha, \quad c^* = 1 - u^* - i^* - m^* > 0.$$

Positive consumption is a consequence, not an additional assumption.

For given $A_0, N_0 > 0$, construct initial levels as

$$B_0^* = \left[A_0 N_0 (k^*)^{\alpha/\lambda} (u^*)^{\beta/\lambda} m^* \left(\frac{\chi}{g_B^*} \right)^{1/\eta} \right]^{\eta\lambda/\Delta}, \quad (53)$$

$$Y_0^* = A_0 N_0 (k^*)^{\alpha/\lambda} (u^* B_0^*)^{\beta/\lambda}, \quad K_0^* = k^* Y_0^*, \quad (54)$$

$$q_0^* = \frac{m^* Y_0^*}{\eta g_B^* B_0^*}. \quad (55)$$

The first two equations solve the initial production equation and $g_B^* B_0^* = \chi (B_0^* m^* Y_0^*)^\eta$ simultaneously. Let $Y_t^* = Y_0^* e^{g_Y^* t}$, $B_t^* = B_0^* e^{g_B^* t}$, and $q_t^* = q_0^* e^{(g_Y^* - g_B^*) t}$. Set $K = k^* Y^*$, $C = c^* Y^*$, $U = u^* Y^*$, $M = m^* Y^*$, $X = B^* U$, and $L = N$. Determine p_X, r, w, Π from the static equations in Definition 1.

Equations (36) and (37) make production and the research law hold at every date. The chosen q_0^* ensures $M = \eta q \dot{B}$, which is the research FOC. The capital-return, Euler, and resource equations follow from k^*, r^*, c^* . Finally, the costate equation reduces to

$$g_Y^* - g_B^* = r^* - \frac{u^* \eta g_B^*}{m^*} - \eta g_B^*,$$

which follows from (45) and $(1 - \eta)g_B^* = \eta g_Y^*$. All quantities remain finite and positive at every finite date.

Both transversality conditions hold explicitly:

$$e^{-\rho t} \frac{N_t K_t^*}{C_t^*} = \frac{N_0 k^*}{c^*} e^{-(\rho-n)t} \longrightarrow 0, \quad (56)$$

$$D_t q_t^* B_t^* = q_0^* B_0^* e^{-(\rho-n)t} \longrightarrow 0. \quad (57)$$

Here $D_t = e^{-r^* t}$. Discounted developer profit is proportional to $e^{-(\rho-n)t}$. Discounted household utility is $N_0 e^{-(\rho-n)t} [\log(C_0^*/N_0) + (g_Y^* - n)t]$. Both objective integrals are finite because $\rho > n$.

To verify global developer optimality, static optimization at given K_t, A_t, L_t gives

$$\pi_t(B) = \beta(1 - \beta)[K_t^\alpha (A_t L_t)^\lambda]^{1/(1-\beta)} (\beta^2 B)^{\beta/(1-\beta)}. \quad (58)$$

This function is concave in B when $\beta \leq 1/2$. Research technology $F(B, M) = \chi B^\eta M^\eta$ is jointly concave when $\eta \leq 1/2$. Since $q_t > 0$, the reduced Hamiltonian $\pi_t(B) - M + q_t F(B, M)$ is jointly concave. Let $J_T = \int_0^T D_t [\pi_t(B_t) - M_t] dt$. For any admissible alternative, denoted by a tilde, with the same B_0 , the supporting-hyperplane inequality, research FOC, and costate equation give

$$\tilde{J}_T - J_T \leq -D_T q_T (\tilde{B}_T - B_T) \leq D_T q_T B_T \longrightarrow 0. \quad (59)$$

No alternative can have a larger limiting payoff, or a larger upper limit of truncated payoff, than the constructed finite-valued candidate. Static optimality was global by strict concavity of (15), so reducing the developer problem did not discard a profitable service choice.

For the household, write $\Lambda_t = e^{-\rho t} N_t / C_t$. Euler implies $\dot{\Lambda} = -r\Lambda$. Concavity of log utility and the linear household budget yield, for any feasible alternative at the same given prices and distributions and with the same K_0 ,

$$\int_0^T e^{-\rho t} N_t [\log(\tilde{C}_t / N_t) - \log(C_t / N_t)] dt \leq -\Lambda_T (\tilde{K}_T - K_T) \leq \Lambda_T K_T \longrightarrow 0. \quad (60)$$

This proves global household optimality. Final-good firms face a concave constant-returns technology, so their first-order conditions maximize profit. Market clearing and both global agent optima establish the claimed equilibrium. $\square$

References

- Feenstra, R. C., Inklaar, R., and Timmer, M. P. (2025). Penn world table version 11.0. Groningen Growth and Development Centre.
- Fonteneau, F., Mollins, J., Marchi, S., Russo, L., Gentaz, A., Daoud, M., and André, A.-A. (2025). Advancing the measurement of investments in artificial intelligence. Technical Report 47, OECD Publishing, Paris.
- OECD (2025). OECD Global Long-Run Economic Scenarios: 2025 Update. Technical Report 36, OECD Publishing, Paris.
- Pardo, O. (2026). The future of growth and human labor under recursive AI self-improvement. [Working paper](#).
- United Nations (2024). World Population Prospects 2024: Summary of Results. Technical Report UN DESA/POP/2024/TR/NO. 9, United Nations, Department of Economic and Social Affairs, Population Division, New York.